

ELECTRIC AND OPTICAL FIBRE CABLES – TEST METHODS FOR NON-METALLIC MATERIALS –

Part 502: Mechanical tests – Shrinkage test for insulations

1 Scope

This Part 502 of IEC 60811 gives the test method for the shrinkage for insulations.

2 Normative references

The following documents, in whole or in part, are normatively referenced in this document and are indispensable for its application. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60811-100:2012, *Electric and optical fibre cables – Test methods for non-metallic materials – Part 100: General*

3 Terms and definitions

For the purposes of this document, the terms and definitions given in IEC 60811-100 apply.

4 Test method

4.1 General

This part of IEC 60811 shall be used in conjunction with IEC 60811-100.

4.2 Test equipment

The test equipment shall be as follows:

- an air oven, electrically heated, capable of maintaining the test temperature;
- a measuring device with a scale division of 0,5 mm.

4.3 Sampling

One sample about 1,5 L mm in length of each core to be tested shall be taken at least 0,5 m away from the end of the cable length.

L is the length given in the relevant cable standard.

4.4 Preparation of test pieces

All coverings, except adherent, extruded semi-conducting screens, if any, shall be removed from the samples of insulated conductor.

Within an interval of not more than 5 min from the time of cutting the samples, a test length of $L \pm 5$ mm shall be marked on the middle part of each piece of core. The distance between the marks shall be measured to an accuracy of 0,5 mm. Each test piece shall then be

prepared by cutting and removing the insulation from both ends of each sample up to positions between 2 mm and 5 mm away from the marks.

4.5 Test procedure

Each test piece shall be supported horizontally in the air oven preheated to the temperature specified in the cable standard by the bare ends of conductors or on the surface of a preheated talc bath, to permit free movement of the insulation. Each test piece should remain in the oven for the time specified in the cable standard.

Each test piece shall then be removed from the oven and allowed to cool in air to room temperature and the distance between the two marks on each piece measured again to the nearest 0,5 mm.

4.6 Expression of results

The percentage shrinkage ΔL is the difference between the distances between the marks before the heat treatment (L_1) and after the heating and cooling (L_2), recorded as a percentage of the distance between the marks before the treatment:

$$\Delta L = \frac{L_1 - L_2}{L_1} \times 100 \%$$

5 Test report

The test report shall be in accordance with that given in IEC 60811-100.